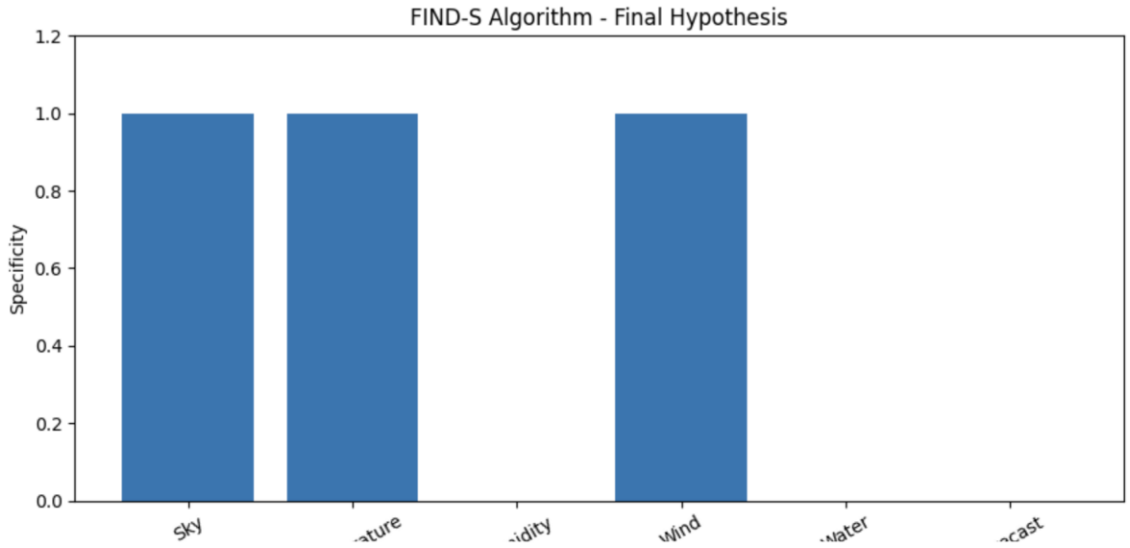
Exp1

```
Sky : Sunny
Temperature : Warm
Humidity : ?
Wind : Strong
Water : ?
Forecast : ?

Final Hypothesis:
<Sunny, Warm, ?, Strong, ?, ?>
```

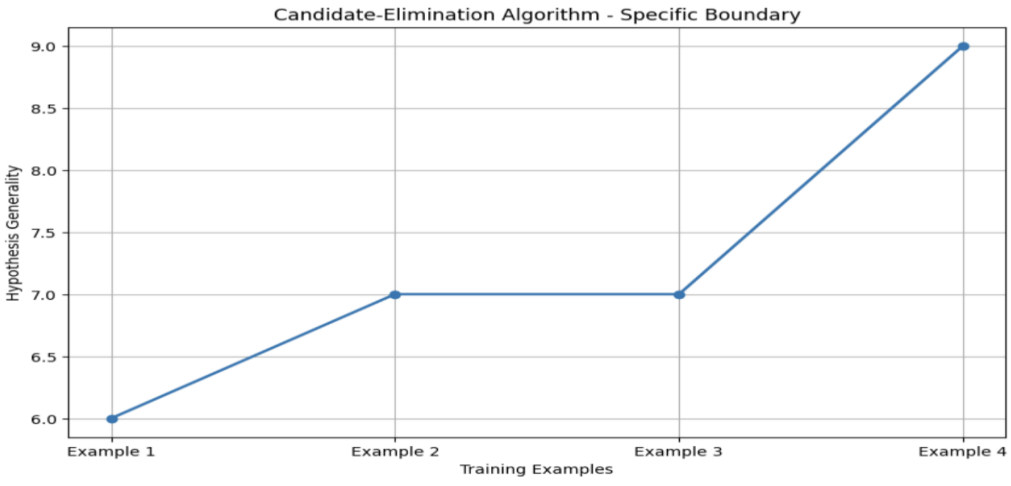


Exp 2

```
TRAINING DATA
     Sky Temperature Humidity    Wind Water Forecast EnjoySport
0  Sunny        Warm   Normal  Strong  Warm     Same        Yes
1  Sunny        Warm     High  Strong  Warm     Same        Yes
2  Rainy        Cold     High  Strong  Warm   Change         No
3  Sunny        Warm     High  Strong  Cool   Change        Yes

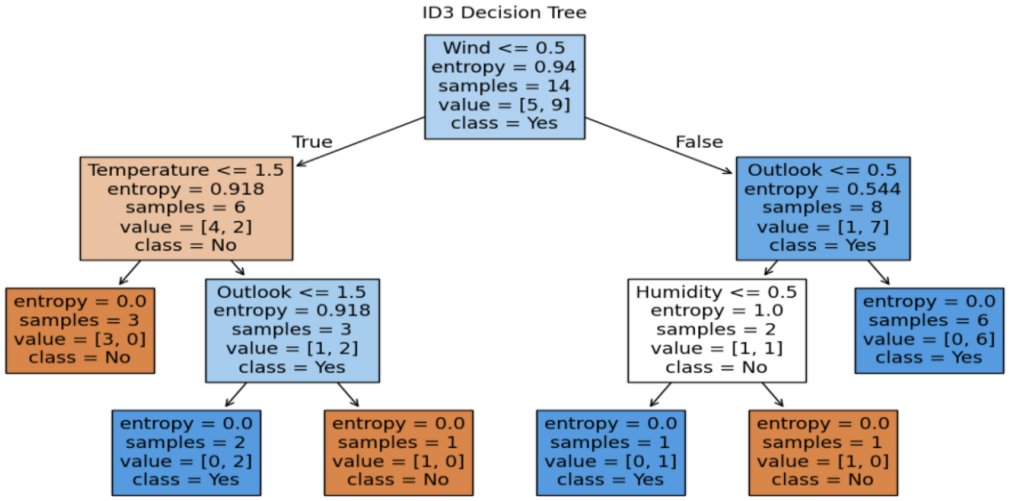
FINAL SPECIFIC BOUNDARY:
['Sunny', 'Warm', '?', 'Strong', '?', '?']

FINAL GENERAL BOUNDARY:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']
```



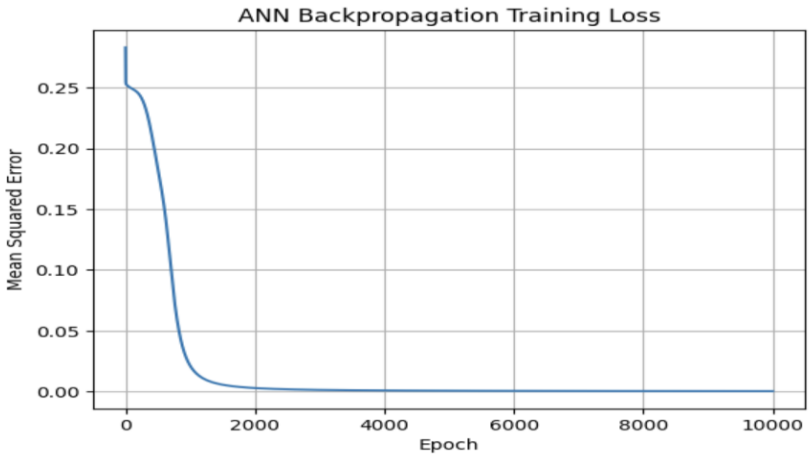
Exp3

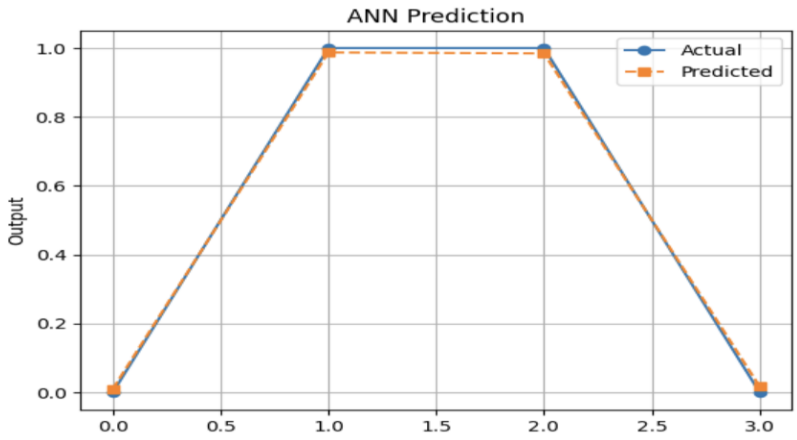
```
Prediction: No
```



Exp4

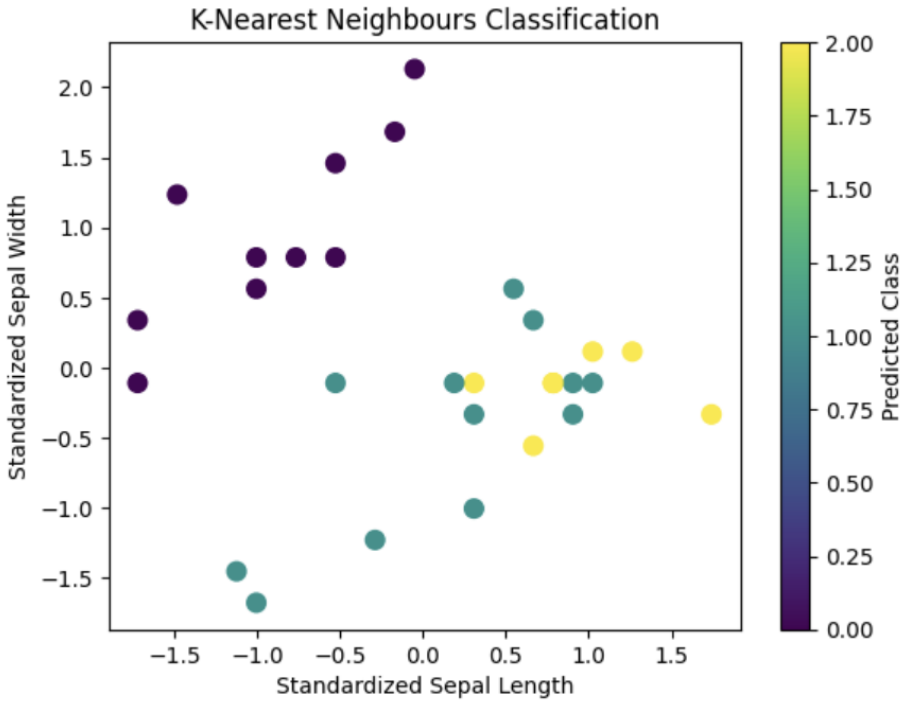
```
Predicted Classes:
[0 1 1 0]
```





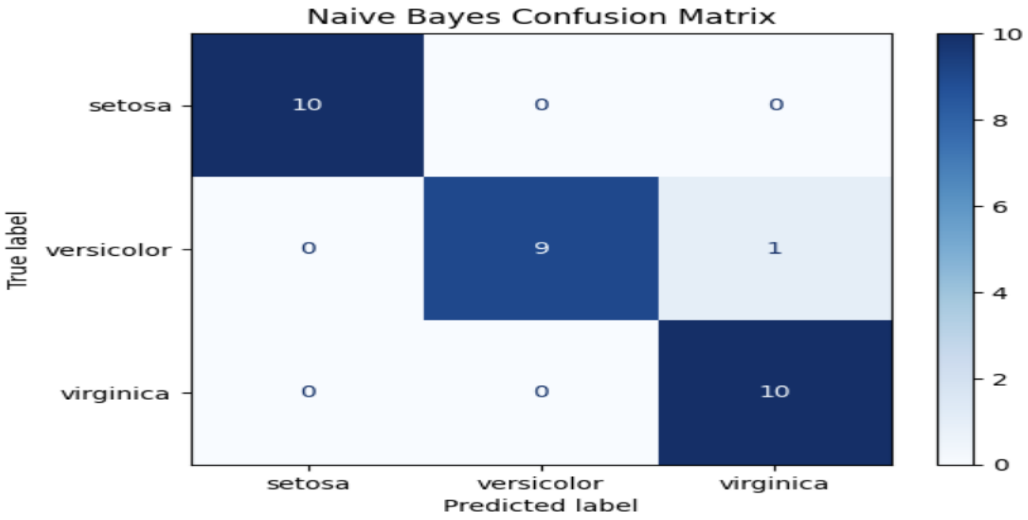
Exp5

```
Actual Classes: [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Predicted Classes: [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 1 0 2 1 1 2 1 1 0 2 0]
Accuracy: 0.9333333333333333
```



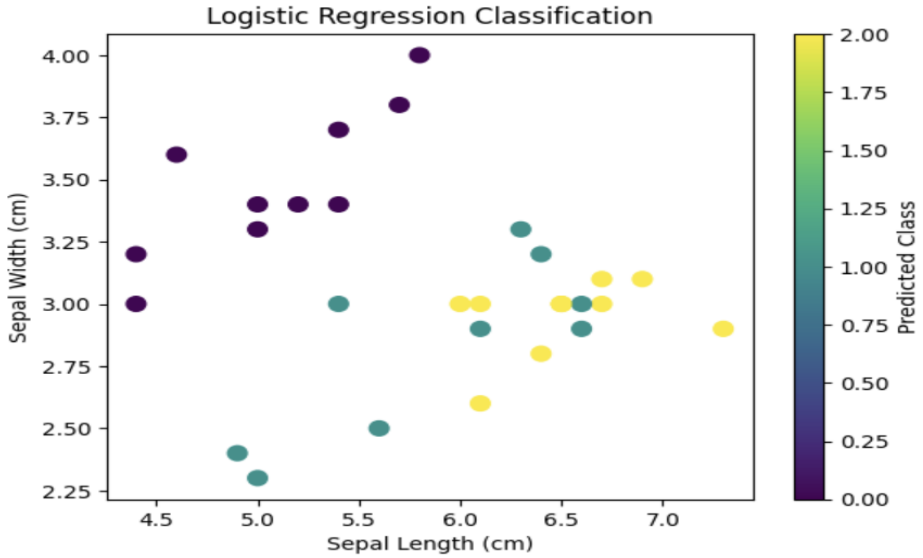
Exp6

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
Accuracy: 0.9666666666666667
```



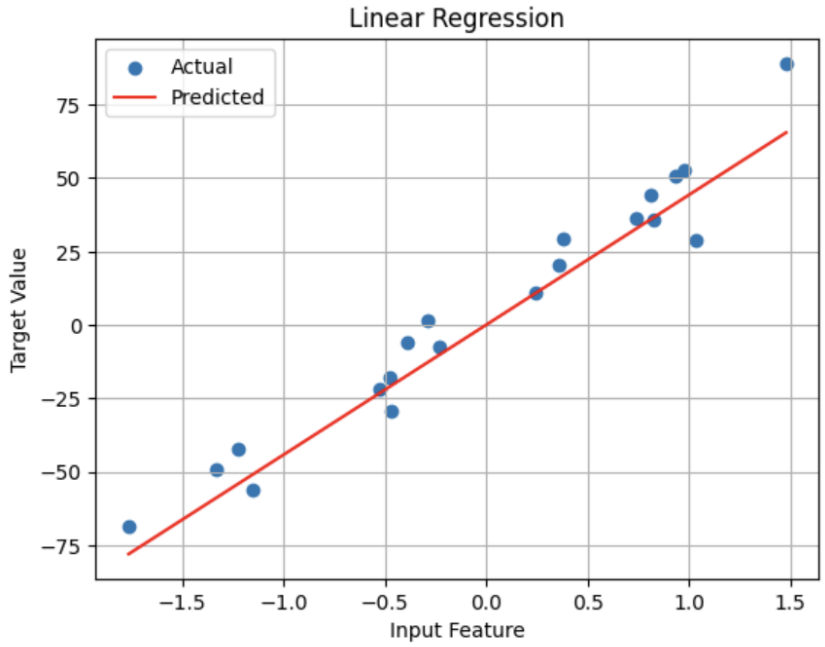
Exp7

```
Actual Classes: [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 1 1 0 2 0]
Predicted Classes: [0 2 1 1 0 1 0 0 2 1 2 2 2 1 0 0 0 1 1 2 0 2 1 2 2 2 1 0 2 0]
Accuracy: 0.9666666666666667
```



Exp8

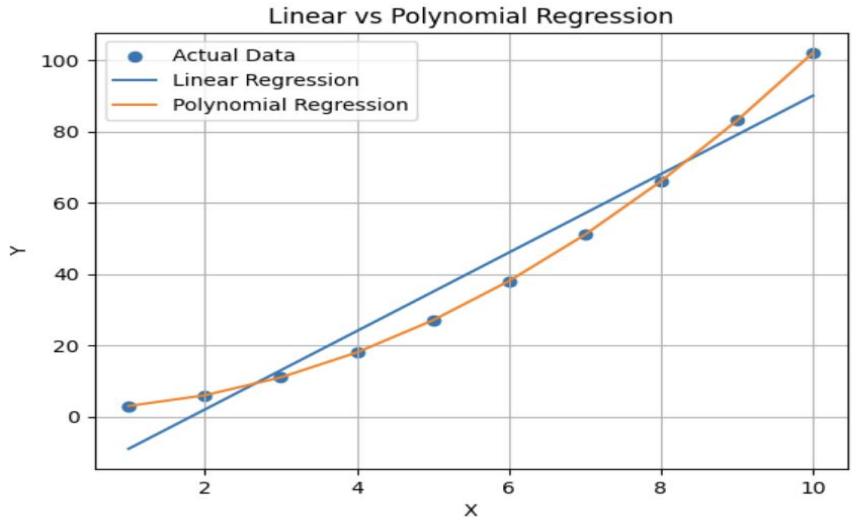
```
Coefficient: 44.24418215597419
Intercept: 0.09922221422587718
Mean Squared Error: 104.20222653187027
R2 Score: 0.9374151607623286
```



Exp9

```
Linear Regression
MSE: 52.79999999999997
R2 Score: 0.9497645211930926

Polynomial Regression
MSE: 2.3922206950827182e-29
R2 Score: 1.0
```



Exp10

```
First 10 Cluster Labels:
[1 1 1 1 1 1 1 1 1 1]
Number of Clusters: 3
Converged: True
Log Likelihood: -1.2013110850984177
```

